

Nested Clustered Optimization (NCO) & Hierarchical Risk Parity (HRP)

Modern Machine Learning Frameworks for Robust Asset Allocation

AI Finance Seminar
Sogang University

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Reference: Marcos M. López de Prado (*Book 2: Machine Learning for Asset Managers*, Ch. 7) & skfolio

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Chapter 7 Overview: Why Cluster-Based Asset Allocation?

Markowitz's Curse (The Root Cause)

- **Matrix Inversion Explosion:** Classical Mean-Variance Optimization (MVO) requires computing V^{-1} .
- **Condition Number** $\kappa(V) \rightarrow \infty$: High cross-asset correlation causes V to become singular ($|V| \approx 0$).
- **The Correlation Paradox:** The more correlated assets are, the greater the need for diversification, yet the more ill-conditioned V^{-1} becomes.

The Machine Learning Paradigm Shift

- **Abandon Fully-Connected Graphs:** Classical optimizers evaluate all $N(N-1)/2$ correlations simultaneously, letting noise contaminate every weight.
- **Impose Tree/Cluster Structure:** Group collinear assets into clusters so optimization occurs within peer groups first.
- **Two Prevailing ML Approaches:**
 - ① **HRP:** Bypasses V^{-1} entirely using recursive tree bisection.
 - ② **NCO:** A 2-stage wrapper that isolates condition numbers inside sub-matrices.

Topic 1: NCO Architecture & 2-Stage Allocation

What is NCO? (López de Prado, 2019)

Nested Clustered Optimization (**NCO**) is a modular wrapper framework that solves Markowitz's Curse by splitting the global allocation problem into **two nested optimization stages**.

Step 1: ONC Clustering

- Partitions N assets into K clusters $C_1, \dots, C_K$ using **Optimal Number of Clusters (ONC)**.
- Uses Silhouette t -stats and Random Matrix Theory (RMT) noise filtering.

Step 2a: Intracluster

- For each cluster k , computes intracluster weights w_k using sub-covariance V_k .
- **Key:** Tames high condition numbers inside isolated sub-matrices.

Step 2b: Intercluster

- Aggregates clusters into K meta-assets:
 $V_{\text{inter}} = W'_{\text{intra}} V W_{\text{intra}}$.
- Computes intercluster weights w_{inter} across well-behaved meta-assets.

Final Allocation Formula

$$\mathbf{w}^* = \mathbf{W}_{\text{intra}} \cdot \mathbf{w}_{\text{inter}}^* \quad \text{where } \mathbf{W}_{\text{intra}} = \text{diag}(w_1, \dots, w_K) \in \mathbb{R}^{N \times K}$$

Topic 2: NCO Mathematical Mechanics

Why NCO Tames the Condition Number

- **Intraccluster Condition Number:** $\kappa(V_k) \ll \kappa(V)$ because each cluster k contains a small, homogeneous sub-matrix.
- **Intercluster Condition Number:** $V_{\text{inter}} = W'_{\text{intra}} V W_{\text{intra}}$ is an $K \times K$ matrix where off-diagonal cross-cluster correlations are virtually zero. Thus, $\kappa(V_{\text{inter}}) \approx 1$.
- **Result:** NCO converts an ill-conditioned $N \times N$ matrix inversion into K small, stable sub-inversions plus one trivial $K \times K$ meta-inversion.

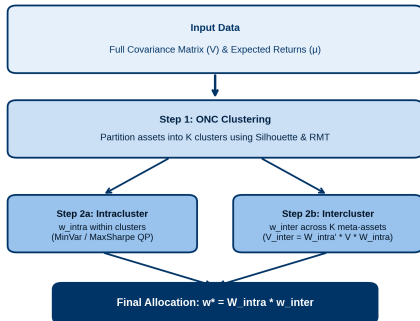
Flexibility of Objective Functions

Unlike HRP (which is strictly risk-based), NCO is an **optimizer-agnostic wrapper**:

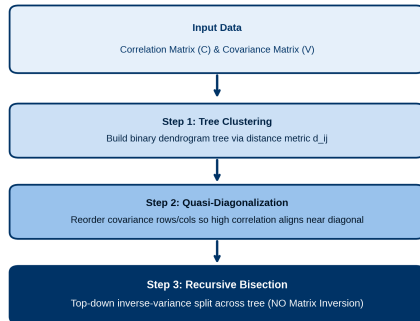
- **NCO Minimum Variance:** $w_k = \frac{V_k^{-1} \mathbf{1}}{\mathbf{1}' V_k^{-1} \mathbf{1}}$ at intra- and inter-levels.
- **NCO Maximum Sharpe Ratio:** Incorporates expected return vector μ : $w_k = \frac{V_k^{-1} \mu_k}{\mathbf{1}' V_k^{-1} \mu_k}$.
- **Constraints:** Supports long-only, max position limits, or Black-Litterman views at both levels.

Topic 3: [Visualization] NCO vs. HRP Architecture

NCO (Nested Clustered Optimization) Architecture



HRP (Hierarchical Risk Parity) Architecture



Architectural Comparison

- **NCO (Left):** Partitions assets via ONC → Solves Quadratic Programs (QP) inside clusters → Solves QP between meta-assets.
- **HRP (Right):** Builds binary dendrogram tree → Reorders matrix (Quasi-Diagonalization) → Recursive inverse-variance top-down bisection (NO QP / NO Matrix Inversion).

Topic 4: Detailed Comparison: HRP vs. NCO

1. Optimization Engine

- **HRP**: Bypasses matrix inversion entirely (V^{-1} not computed). Uses recursive top-down inverse-variance bisection.
- **NCO**: Uses Quadratic Programming (MVO / Max Sharpe) inside and between clusters.

2. Clustering Method

- **HRP**: Hierarchical Agglomerative Clustering (HAC / single linkage) building a continuous binary tree.
- **NCO**: Partitional clustering via ONC (Optimal Number of Clusters) based on Silhouette t -stats.

3. Expected Return (μ) Inclusion

- **HRP**: Purely risk-based (risk parity / inverse variance). Does not incorporate expected returns μ .
- **NCO**: Fully incorporates μ for Max Sharpe Ratio, Black-Litterman, or custom alpha views.

4. Condition Number Taming

- **HRP**: Completely eliminates condition number issues (κ irrelevant).
- **NCO**: Isolates high κ inside cluster sub-matrices V_k .

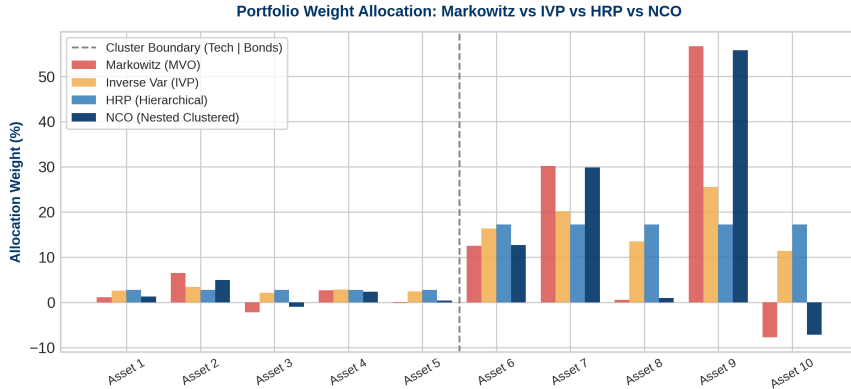
Topic 5: Comparative Summary Matrix

Feature	Markowitz (MVO)	Inverse Var (IVP)	HRP	NCO (Ne)
Matrix Inversion?	Required (V^{-1})	None (Diagonal)	None (Tree Bisection)	Required inside
Condition Risk (κ)	High (Explodes)	Zero (Ignores corr)	Zero (Bypasses V^{-1})	Low (Isolate)
Graph Topology	Fully-connected	Unconnected	Hierarchical Tree	Partitioned
Uses Expected Returns (μ)?	Yes	No	No	Yes
Weight Concentration	Extreme	Uniform	Balanced	Balanced
OOS Performance	Poor (High Variance)	Suboptimal	Robust	Superior

When to Use Which?

- **Use HRP:** When return estimates μ are unavailable/unreliable and you want a pure risk-based allocation immune to matrix singularity.
- **Use NCO:** When you have active return views (μ) or want to optimize specific objectives (Max Sharpe, Min Var) while controlling estimation noise.

Topic 6: [Visualization] Empirical Portfolio Weight Allocation



Weight Distribution Observations

- **Markowitz (Red):** Shows extreme concentration and unstable weights due to correlation within clusters.
- **IVP (Yellow):** Allocates uniformly without recognizing that Tech stocks share common risk.

Topic 7: Empirical Results — Monte Carlo Experiments

Minimum Variance Portfolio (50 Assets)

- Monte Carlo simulation (1,000 trials, 10 block clusters with $\rho_{\text{intra}} = 0.5$):
- **Markowitz Raw RMSE:** 7.95×10^{-3}
- **Ledoit-Wolf Shrinkage RMSE:** 8.89×10^{-3}
- **NCO RMSE:** 4.21×10^{-3}
- **RMSE Reduction:** **47.02% error reduction** from NCO alone!

Maximum Sharpe Ratio Portfolio

- Monte Carlo simulation evaluating optimal tangency allocations:
- **Markowitz Raw RMSE:** 7.02×10^{-2}
- **Ledoit-Wolf Shrinkage RMSE:** 6.54×10^{-2}
- **NCO RMSE:** 3.17×10^{-2}
- **RMSE Reduction:** **54.83% error reduction** by replacing Markowitz with NCO!

Key Finding

Ledoit-Wolf shrinkage shrinks all elements blindly toward a diagonal, diluting true signal. NCO preserves intracluster signals while eliminating cross-cluster noise.

Summary: Key Takeaways on NCO & HRP

Core Principles for Quantitative Portfolio Managers

- 1 **Recognize Markowitz's Curse:** Classical convex optimization fails in high-dimensional correlated asset spaces due to condition number explosion.
- 2 **HRP for Pure Risk Parity:** Use HRP when return expectations μ are noisy or unknown. HRP provides a robust, zero-inversion hierarchical risk allocation.
- 3 **NCO for General Optimization:** Use NCO when active returns μ or specific constraints are required. NCO isolates condition numbers inside clusters while preserving optimal tangency allocations.
- 4 **Modular Synergy:** Both NCO and HRP can be combined with Random Matrix Theory (RMT) denoising and Variation of Information (VI) distance metrics for maximum out-of-sample stability.

Thank You!

Questions & Discussion

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